Linear Algebra Cheatsheet

A one-page refresher rendered with KaTeX.

Vectors & norms

The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is $\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$, and the dot product is $\mathbf{x}^\top \mathbf{y} = \sum_i x_i y_i$.

Matrix identities

$$(AB)^\top = B^\top A^\top \qquad (A^{-1})^\top = (A^\top)^{-1}$$

For a square matrix A, the eigenvalue equation is:

$$A\mathbf{v} = \lambda \mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}$$

Useful decompositions

Name	Form	Notes
Eigen	$A = Q\Lambda Q^{-1}$	square, diagonalizable
SVD	$A = U\Sigma V^\top$	any real matrix
Cholesky	$A = LL^\top$	symmetric positive-definite

Gaussian density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\,|\Sigma|^{1/2}} \exp\left(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$